ELECTRONIC SPECTROSCOPY OF ATOMS

{2018}

5.a] Write down the wave function expression and energy expression for H atom.

Answer:

The wave function expression (Ψn), for the H-atom -

$$\Psi_n = f\left(\frac{r}{a_0}\right) \exp\left(-\frac{r}{na_0}\right)$$

where, $a_0 = \frac{\hbar^2}{4\pi^2 me^2}$

r = radial distance from the nucleous.

$f\left(\frac{r}{a_0}\right)$ = Power series

n = principal quantum number.

Energy expression for the H-atom -

$$E_n = -\frac{me^4}{8h^2\varepsilon_0^2 n^2} \text{ J}$$

$$\varepsilon_n = -\frac{me^4}{8h^3c\varepsilon_0^2 n^2} \text{ cm}^{-1}$$

$$= -\frac{R}{n^2} \text{ cm}^{-1}$$

where, ε_0 = vaccum Permitivity.

R = Rydberg constant.

$$= \frac{me^4}{8h^3c\varepsilon_0^2}$$

b// What are the atomic quantum numbers? Discuss their significance.

Answer:

Atomic quantum numbers:- The principal quantum number, orbital or azimuthal quantum number, magnetic quantum number and spin quantum number are also called atomic quantum number.//

There are four types of atomic quantum number-

Quantum NO.	Allowed values	Function
① Principal quantum number, n	1, 2, 3 ...	Governs the size and energy of the orbital
Orbital or azimuthal l	(n-1), (n-2) 0	Governs the shape of the orbital and the electronic angular momentum.
Magnetic, m	±l, ±(l-1) ... 0	Govern the direction of an orbital and electron behaviour in a magnetic field.

spin (s) $\pm \frac{1}{2}$ Govern axial angular momentum of the electron.

c)) what are the values of L and S for the term symbols $^3S_1$ and $^2P_{3/2}$

Answer: Given, $^3S_1$ and $^2P_{3/2}$.

we know, $^{2S+1}L_J$

$2S+1 = 3 \quad J = 1$

$\Rightarrow S = 1 \quad L = 0$

$^2P_{3/2}$

$2S+1 = 2$

$\Rightarrow S = 1/2 \quad L = 1$

d)) Explain compound doublet spectra with energy diagram and spectrum for the transition betn 2p and 2p levels in the fine H atom spectra.

Answer:

- The selection rule for n and ℓ are

Δn = anything $\Delta \ell = \pm 1$ only.

and $\Delta J = 0, \pm 1$.

These selection rules indicates that transition betn 2p and 2D levels are rather more complex.

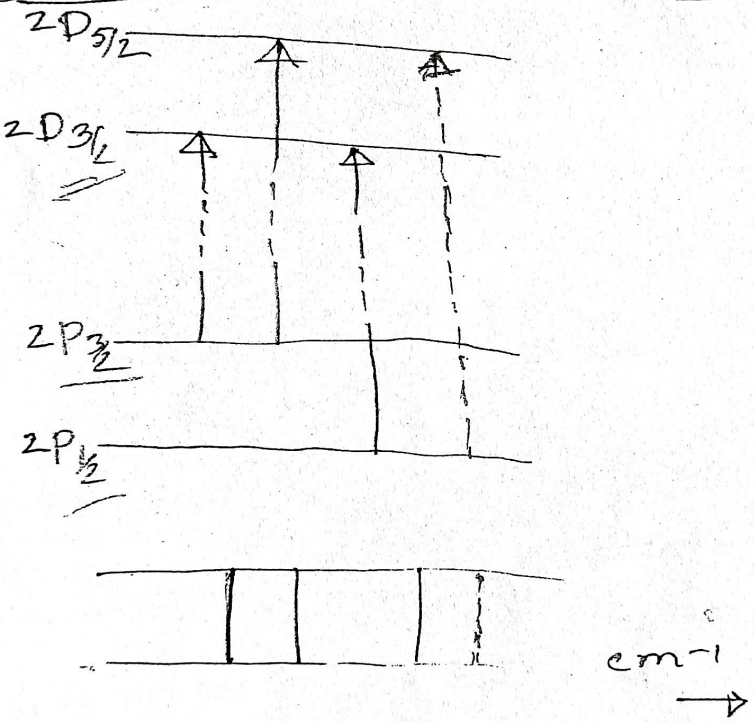


Fig: Compound doublet spectrum for the transition between 2p and 2D states in the H-atom.

The figure shows the transition betn 2p and 2D states, four of the energy levels involved.

The transition at lowest frequency occurred betn the closest levels, the $2p_{3/2}$ and $2D_{3/2}$ and this is allowed, $\Delta J = 0$.

The next transition $2p_{3/2} \rightarrow 2D_{5/2}$ is also allowed because, the value of $\Delta J = +1$.

Thirdly, $2p_{1/2} \rightarrow 2D_{3/2}$ also allowed because of the value of $\Delta J = +1$.

But, the fourth transition (shown dotted) $2p_{1/2} \rightarrow 2D_{5/2}$ is not allowed, since for this $\Delta j = +2$

These spectrum will consist of three lines shown at the foot of the figure. This spectrum is a "compound doublet".

(ii) sketch the energy level of lithium atom in a figure showing the difference in energy of s, p and d states with the same value of n and also show some allowed transition in the figure.

Answer:

The electronic configuration of Li is $1s^2 2s^1$. The alkali metal as Li, Na, K, Rb all have a single electron in the outside shell.

The selection rules for alkali metal is same as the H-atom, that is—

Δn = anything $\Delta l = \pm 1$ $\Delta j = 0, \pm 1$.

Thus transition from the ground state ($1s^2 2s$) can occur to p levels.

$2S_{1/2} \rightarrow nP_{1/2, 3/2}$. a series of doublet similiar to the Lyman series will be formed.

Now, the another transition also occured from 2p state and two separate series of lines will be seen.

$2^2P_{1/2, 3/2} \rightarrow n\,^2S_{1/2}$ (Doublet)

$2^2P_{1/2, 3/2} \rightarrow n\,^2D_{3/2, 5/2}$ (compound doublet)

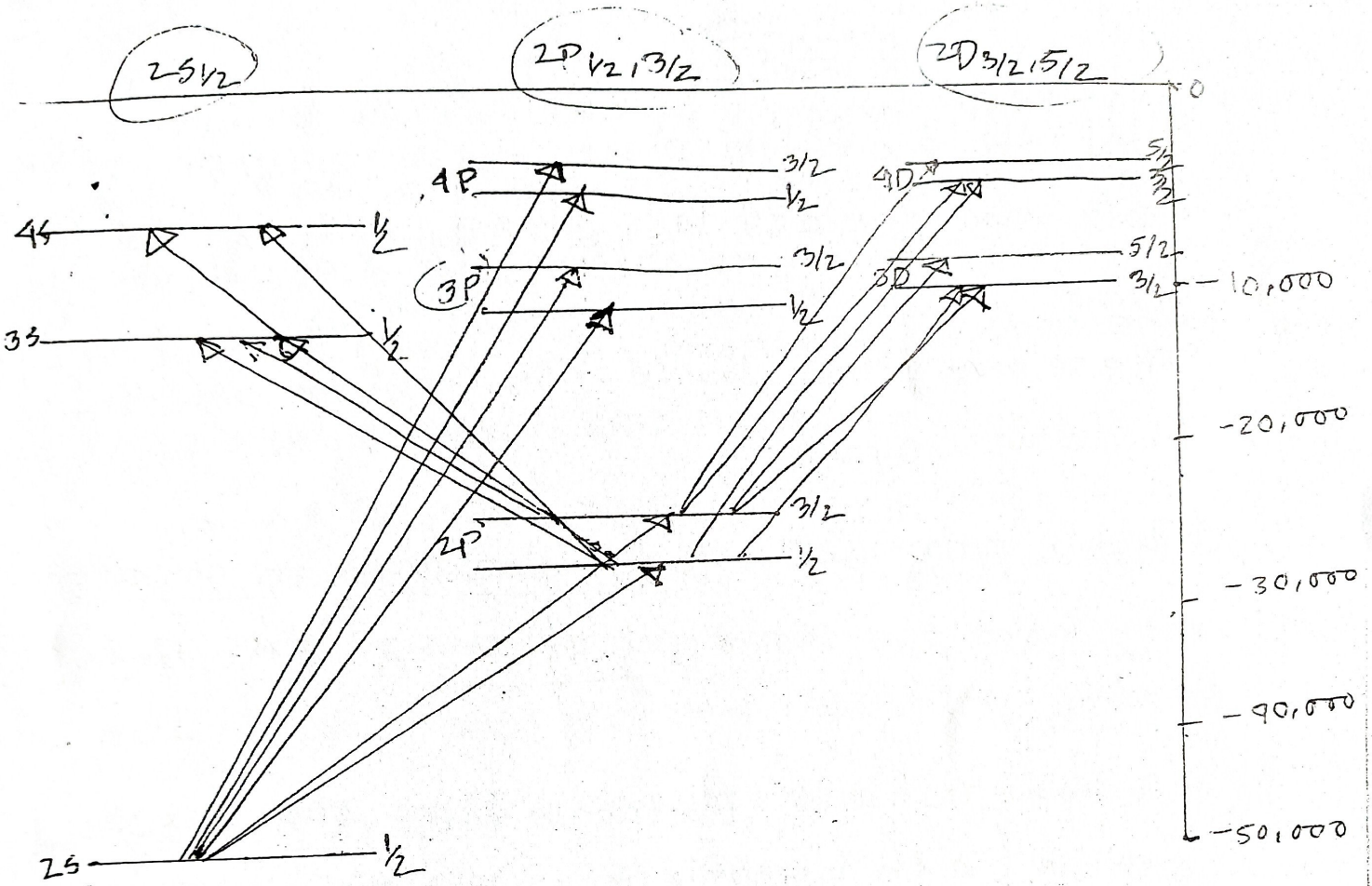


Fig: The energy levels of lithium atom in a figure showing the difference in energy of s, p and d states with the same value of n and transition.

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6| a| State and explain building up principle for many electron atom with example.

Answer:

There are three basic rule, known as the building up rules, —

① Pauli's principle :- No two electrons in an atom may have the same set of values for n, l, m and s_z.

2. Aufbau principle : Electron's tend to occupy the orbital with lowest energy available.

3. Hund's rule : Electrons tend to occupy degenerate orbital singly with their spins parallel.

Explanation:

② The first principle said that, if any atom contain two electrons, then the value of n, l and m may be the same, but their spin must be different.

consider a example — He ($1s^2$)

For first electron — $1s^1$ - $n=1$ $l=0$ $m=0$ $s=+\frac{1}{2}$

2nd " - $n=1$ $l=0$ $m=0$ $s=-\frac{1}{2}$

spin is different.

2. According to this rule, electron first occupies in the lower energies orbital and then higher energies orbital. Orbital energies depends on the value of $(n+l)$. With increasing the value of n and l, orbital energies also increased.

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s$ ----------

(3) According to the Hund's rule, electron first occupy $2p_x$ then $2p_y$ and $2p_z$ ~~both~~ all orbitals are de-generate and spins are parallel if any electron present, then they are paired.

$N(2p^3)$ = ↑ | ↑ | ↑
$2p_x$ $2p_y$ $2p_z$

$O(\otimes 2p^4)$ = ↑↓ | ↑ | ↑
$2p_x$ $2p_y$ $2p_z$

b (i) Sketch the electronic energy levels and show transition between them for single electron of H-atom.

$^4P_{3/2}$, $^2D_{5/2}$

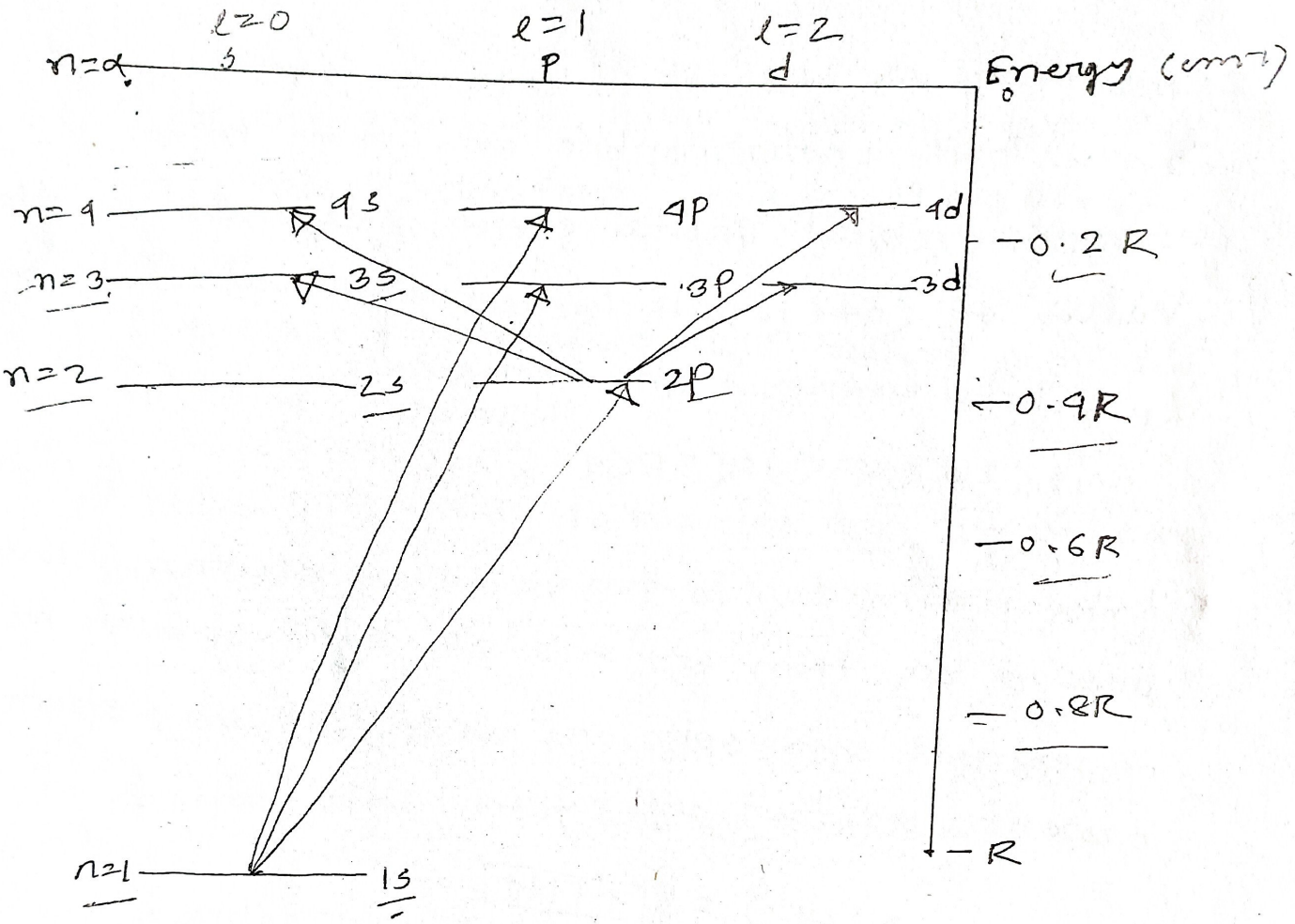


Fig: electronic energy levels and transition betn them for the single electron of the H-atom.

(ii) state the selection rule for H atom spectrum. which of the following transition are allowed for H-atom spectrum?

Answer:

The selection rule for H-atom is.

Δn = anything $\Delta l = \pm 1$.

$2p \rightarrow 4d$

Δn = anything = -2

$\Delta l = +1$ so, transition allowed.

$3p \rightarrow 4s$

Δn = anything = +1

$\Delta l = -1$ so, transition allowed.

$2p \rightarrow 4f$

Δn = anything

$\Delta l = +2$, so, transition not allowed.

$2p \rightarrow 1s$ Δn = anything.

$\Delta l = -1$ transition allowed.

(iii) show with spectrum that the liner of Lymen series are observed at 3R/R, 8R/9, 15R/16 ..... cm^{-1} in the H atom speetrum.

Answer: The electronic configuration of H atom (H(1)) = $1s^1$.

The selection rule for H atom Δn = anything

$\Delta l = \pm 1$ only.

So, the electron in the ground state 1s can undergo transition into p state.

$1s \rightarrow np$.

then, 2p electron also undergoes transition into s or d state.

$2p \rightarrow ns$ or nd.

In general, an electron in a lower state n'' can undergo in a higher state n', with absorption of energy.

$$\Delta\varepsilon = \varepsilon_{n'} - \varepsilon_{n''} \text{ cm}^{-1}.$$

$$\bar{\nu}_{spect} = -\frac{R}{n'^2} - \left(-\frac{R}{n''^2}\right)$$

$$= R\left(\frac{1}{n''^2} - \frac{1}{n'^2}\right) \text{ cm}^{-1}.$$

For transition: (Lyman series)

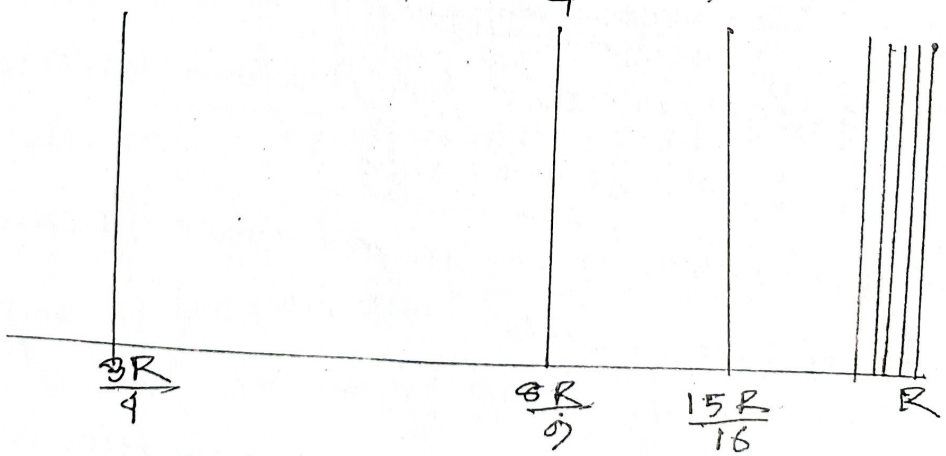
$1s \longrightarrow np$

$$\bar{\nu}_{Lyman} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right) \text{ cm}^{-1}$$

$n = 2, 3, 4 \ldots$

$$= R - \frac{R}{n^2} \text{ cm}^{-1}$$

$$= \frac{3R}{4}, \frac{8R}{9}, \frac{15R}{16} \text{ ---- cm}^{-1}$$



Q1 (i) What is the selection rule for many electron system in case of electronic spectroscopy.

(ii) Discuss the transition from 3p to 3p states for He atom spectrum with necessary energy diagram and spectrum. Why the spectrum obtained from 3p → 3p transition is reffered to as compound triplet.

Answer:

The electronic configuration of He is

$He(2) = 1s^2$.

Two spins of the electron are paired.

$s_z = +\frac{1}{2} - \frac{1}{2} = 0$, singlet state

$2s+1 = 1$

spins are parallel, $s_z = 1$, and triplet state

$2s+1$

$= 3$.

Selection rules for many-electron systems are –

$\Delta S = 0$, $\Delta L = \pm 1$ $\Delta J = 0, \pm 1$

For singlet state $s = \frac{1}{2}$

Initially $^1s_0$ state $L=0, S=0, J=0$ can undergo a transition to np^1 state, where $L=1$, $S=0$, $J=1$.

$^1s_0 \longrightarrow {}^1p_1$

. Then $^1P_1$ undergoes another transition to $^1S_0$ or $^1D_2$ states. where the value of $S=0, L=2, J=2$ and the spectral line will be singlet.

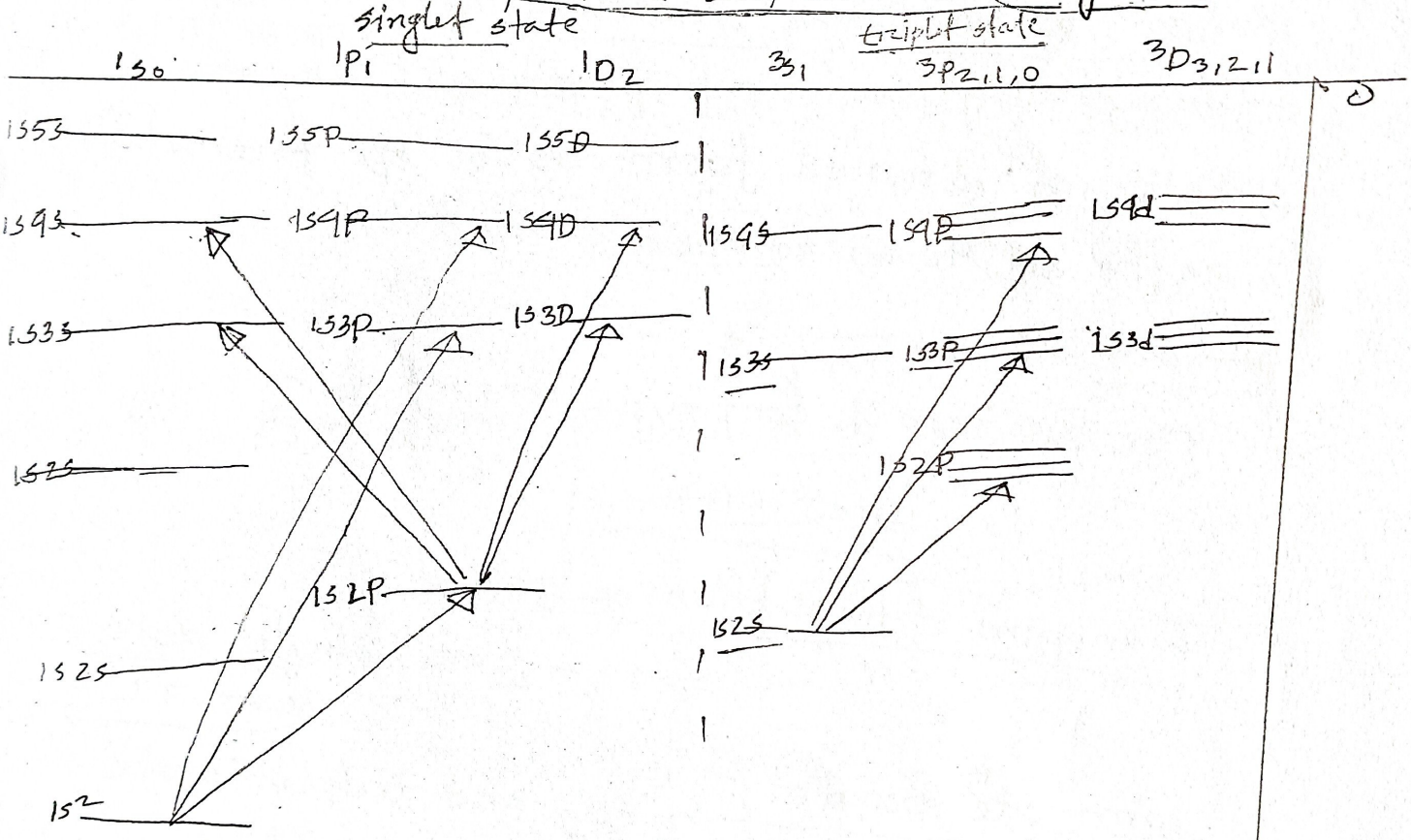


Fig: Some energy levels of the electron in the He atom, with allowed transition.

For parallel electron, $S=1, L=0$ $J=1$ and $3s_1$ can undergo transition into (1s1p) triplet state or $n^3P_{2,1,0}$ where, $S=1$ $L=1$ $J=2,1$ or 0,

so,

$$3s_1 \longrightarrow 3P_2, 3P_1, 3P_0$$

All three transition are allowed since $\Delta J = 0, \pm 1$ and spectral lines will be triplets.

Then transition from 3p state to triplate $3s$ state and ^3D state also occur. For ^3D, $S=1, L=2$ $J=3,2,1$ so, the transition betn 3p and 3p staten-.

$$^3P_2 \longrightarrow {}^3D_{3,2,1}$$

$$^3P_1 \longrightarrow {}^3D_{2,1}$$

$$^3P_0 \longrightarrow {}^3D_1$$

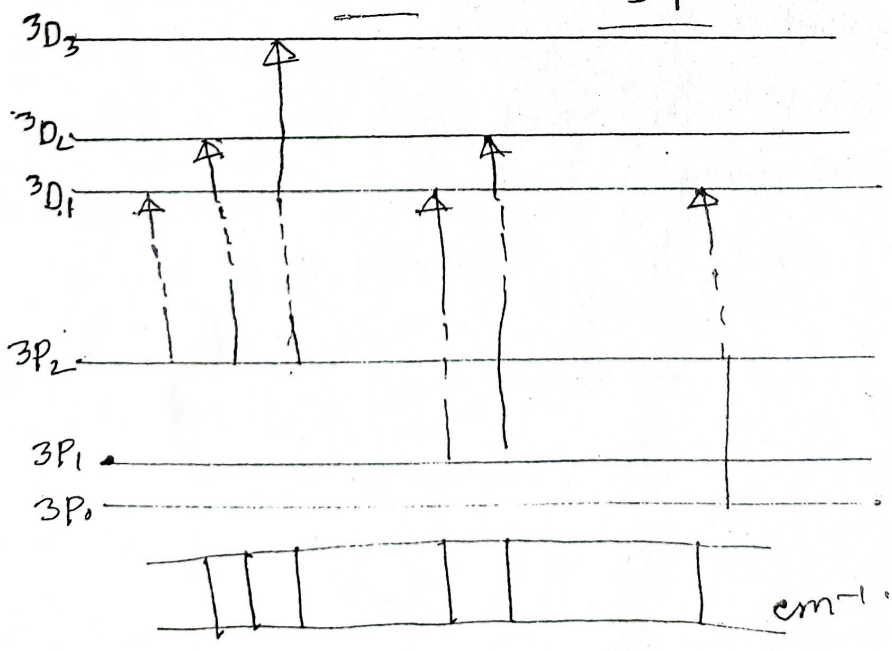


Fig: Compound triplet spectrum for the transition bet 3p and 3p staten.

d) Find the values of L, S and J for $^4P_{3/2}$ and $^2D_{5/2}$.

Answer:

We know, Term symbol = $^{2S+1}L_J$

$^4P_{3/2}$

$2S+1 = 4$

$\Rightarrow S = \frac{3}{2}$

$L = 1$ for P

$J = L+S-1$

$= \frac{3}{2}+1-1$

$= \frac{3}{2}$.

$^2D_{5/2}$

$2S+1 = 2$.

$\Rightarrow S = \frac{1}{2}$

$L = 2$

$J = L+S$

$= 2+\frac{1}{2}$

$= \frac{5}{2}$

(e) What are meant by H and He like species give examples.

Answer:

Any atom which contain a single electron, in their outer shell and their spectrum like hydrogen atom.

The He^+, Be^+, B^{2+} ions have one electron in their outside shell like an H-atom and their spectrum will be similiar to the H atom.

$H(1) = 1s^1$

$He^+(1) = 1s^1$

$Be^+(3) = 1s^2 2s^1$

$B^{2+}(3) = 1s^2 2s^1$

Any atom which contain two electrons in their outer shell and the spectrum will be similiar to the He atom.

The B^+, C^{2+} ions have two electrons in their outer shell and their spectrum will be similiar to the He- atom.

$He(2) = 1s^2$

$B^+(4) = 1s^2 2s^2$

$C^{2+}(4) = 1s^2 2s^2$.

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5 (b) How can you get the term symbol for the following ?

(i) $L=1$, $S=1$ and $J=2$

$L=2 \quad S=\frac{1}{2} \quad J=\frac{3}{2}$.

$^{2S+1}L_J$

(i) $S=1 \quad 2.1+1=3$

$L=1 = P$

$J=2$

$^3P_2$

(ii) $2.S+1 = 1+1=2. \quad L=2=D$

$^2D_{3/2}$.

(c) What is meant by j-splitting?

Answer :-

There is no attempt to show the energy level splitting of the p and D states, the separation betn these levels differing only in j. The difference betn j values is termed as j-splitting.

selection rules for the J value is $\Delta j = 0, \pm 1$.
j-splitting is many thousands times smaller than the n value.

Q] Explain why the liner in atomic spectra of Na are doublet.

Answer:

The electronic configuration of Na is

$Na(11) = 1s^2\ 2s^2\ 2p^6\ 3s^1$

~~Here L=~~

At excited state, electron goes from 3s to 3p orbital.

Now, $L = 1$ and $S = 1/2$.

multiplicity $= 2S+1 = 2$.

$J = L+S, L+S-1$

$= 1+1/2,\ 1+1/2-1$

$= 3/2,\ 1/2$.

Term symbol $= {}^{2S+1}L_J$

$= {}^2P_{3/2},\ {}^2P_{1/2}$.

Here, we get two term symbol.
So, Na atom show doublet spectra.

** ① Russell-saunders coupling:

First sum of the orbital contribution, then the spin contribution separately and finally add the total spin and total orbital contribution to reach the grand total.

$$\Sigma L_i = L \quad \Sigma s_i = S \quad L + S = J$$

② J-j coupling:-

sum of the orbital and spin momentat of each electron separately, finally summing the individual total to form grand total.

$$l_i + s_i = j_i \quad \Sigma j_i = J$$

Applications of Atomic absorptim spectroscopy:

(i) Water analysis (Ca, Mg, Fe, Ag)

(ii) food analysis: analysis of animal feedstuff (Mn, Fe, Cu, Cr, Zn)

(iii) analysis of soil

(iv) clinical analysis (blood samples, whole blood, plasma, serum, Ca, Mg, Fe, Na, K)

(v) Analysis of additives in lubricating oils and greases (Ba, Ca, Na, Zn, Mg)

THE END

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